



APPOINTMENT MANAGEMENT SYSTEM

SYSTEM ARCHITECTURE

USERS



PATIENT / USER

- Register / Login
- Book Appointment
- View / Reschedule
- Cancel Appointment
- View History



DOCTOR

- View Appointments
- Manage Schedule
- Update Availability
- View Patient Details



ADMIN

- Manage Users
- Manage Doctors
- Manage Departments
- Manage Appointments
- View Reports



RECEPTIONIST

- Book Appointment
- Manage Appointments
- Check Doctor Availability
- Generate Reports



ACCESS

- Web Application
- Desktop Application
- (Future) Mobile App

APPLICATION LAYER

APPLICATION LAYER (CORE JAVA)

User Management Module



- Registration
- Login / Logout
- Profile Management
- Role Management

Appointment Management Module



- Book Appointment
- View Appointments
- Reschedule
- Cancel Appointment

Doctor Management Module



- Add / Update Doctor
- Specialization
- Manage Availability
- Doctor Schedule

Schedule Management Module



- Set Availability
- Time Slots
- Break Management
- Holiday Management

Notification Module



- Email Notifications
- SMS Notifications
- Appointment Reminders
- Alerts

Report & Analytics Module



- Appointment Reports
- Doctor-wise Reports
- Daily / Monthly Reports
- Export Reports

System Management Module



- Settings
- Backup & Restore
- Activity Logs
- Data Management

APPIA LAYER

FILE STORAGE (DATA FILES)



- | | |
|-------------------|-----------------------|
| users.dat | (User Information) |
| doctors.dat | (Doctor Information) |
| appointments.dat | (Appointment Details) |
| schedule.dat | (Doctor Schedule) |
| notifications.dat | (Notification Logs) |
| settings.dat | (System Settings) |

DATA HANDLING



- File Read / Write
- Data Validation
- Exception Handling
- Data Integrity

EXTERNAL SERVICES (OPTIONAL / FUTURE)



Email Service (Send Emails)



SMS Gateway (Send SMS)



Payment Gateway (Future)



Cloud Storage (Backup)



REPORTS & DASHBOARD (POWER BI)



Total Appointments



Appointments by Doctor



Daily / Monthly Trends



Department-wise Report



Booking Source Analysis



Export & Share Reports

SYSTEM FLOW



1. User Books Appointment



2. System Checks Availability



3. Appointment Confirmed



4. Notification Sent to User



5. Doctor Views Appointment



6. Reports & Insights Generated

BENEFITS



Efficient Appointment Scheduling



Reduced Waiting Time



Better Resource Utilization



Automated Notifications



Real-time Reports & Insights



Improved Patient Satisfaction



Secure & Reliable Data Management